

EC400 Probability and Statistical Inference

Final Exam Exam Solutions September 2026

If you feel that you need to make an additional assumption, state it clearly before proceeding.

Question 1

Let W describe whether a computer system is working: $W = 1$ if it is working and $W = 0$ if it is not. Let $P(W = 0) = \pi$, where $0 < \pi < 1$. There are two monitoring devices. Let A be the event that the first device sounds its alarm, and let B be the event that the second device sounds its alarm. Write

$$P(A | W = 1) = a_1, \quad P(A | W = 0) = a_0, \quad P(B | W = 1) = b_1, \quad P(B | W = 0) = b_0,$$

where $0 < a_1 < a_0 < 1$ and $0 < b_1 < b_0 < 1$. Assume that the alarms are independent conditional on $W = 0$, and also independent conditional on $W = 1$. For an observed event C for which the following ratio is defined, let

$$\text{odds that the system does not work after observing } C := \frac{P(W = 0 | C)}{P(W = 1 | C)}.$$

[25 marks]

- (a) Write each of the following in terms of π , a_0 and a_1 : $P(A)$, $P(W = 0 | A)$, and the odds that the system does not work after observing A . [5 marks]

Solution: The law of total probability gives

$$P(A) = \pi a_0 + (1 - \pi) a_1.$$

Bayes' rule therefore gives

$$P(W = 0 | A) = \frac{\pi a_0}{\pi a_0 + (1 - \pi) a_1}.$$

Dividing this probability by the probability that the system is working gives

$$\frac{P(W = 0 | A)}{P(W = 1 | A)} = \frac{\pi}{1 - \pi} \frac{a_0}{a_1}.$$

Thus the odds after observing the alarm equal the initial odds multiplied by a_0/a_1 , the ratio of the alarm probabilities in the two states.

Marking guide: 1 mark for total probability; 2 for Bayes' rule; 2 for the odds formula.

- (b) Show that observing A increases the probability that the system is not working. Explain how the assumption $a_0 > a_1$ leads to this result. [4 marks]

Solution: Since all denominators are positive,

$$\begin{aligned} P(W = 0 \mid A) > \pi &\iff \frac{\pi a_0}{\pi a_0 + (1 - \pi)a_1} > \pi \\ &\iff a_0 > \pi a_0 + (1 - \pi)a_1 \\ &\iff a_0 > a_1, \end{aligned}$$

which holds by assumption. Equivalently, observing the alarm multiplies the odds by $a_0/a_1 > 1$. In words, an alarm is evidence that the system is not working only if the alarm is more likely when the system is not working than when it is working normally.

Marking guide: 2 marks for showing that the probability increases; 2 for explaining the role of $a_0 > a_1$.

- (c) Derive the odds that the system does not work after observing both A and B . Identify exactly where conditional independence is used. **[5 marks]**

Solution: Conditional independence gives

$$P(A \cap B \mid W = 0) = a_0 b_0, \quad P(A \cap B \mid W = 1) = a_1 b_1.$$

Bayes' rule in odds form then yields

$$\frac{P(W = 0 \mid A \cap B)}{P(W = 1 \mid A \cap B)} = \frac{\pi}{1 - \pi} \frac{a_0}{a_1} \frac{b_0}{b_1}.$$

Conditional independence is used only to write each joint conditional probability as a product. It does not imply that the alarms are independent without conditioning on whether the system is working.

Marking guide: 2 marks for the factored conditional likelihoods; 2 for the posterior-odds formula; 1 for locating and qualifying the independence assumption.

- (d) Let I_A and I_B be the indicator (dummy) variables for the events A and B : $I_A = 1$ if A occurs and $I_A = 0$ otherwise, with I_B defined similarly. Prove that

$$\text{Cov}(I_A, I_B) = \pi(1 - \pi)(a_0 - a_1)(b_0 - b_1).$$

Are the alarms independent (without conditioning on W)? Give some intuition for why or why not, noting that they are independent conditional on W . *Hint:* note that $\mathbb{E}[I_A] = 1 \times P(A) + 0 \times P(A^c) = P(A)$. As such, start by writing $\mathbb{E}[I_A I_B]$ as a probability. **[6 marks]**

Solution: Conditional independence and total probability give

$$\mathbb{E}[I_A I_B] = P(A \cap B) = \pi a_0 b_0 + (1 - \pi) a_1 b_1.$$

Also

$$\mathbb{E}[I_A] = \pi a_0 + (1 - \pi) a_1, \quad \mathbb{E}[I_B] = \pi b_0 + (1 - \pi) b_1.$$

Expanding their product and subtracting it from the joint probability gives

$$\text{Cov}(I_A, I_B) = \pi(1 - \pi)(a_0 - a_1)(b_0 - b_1) > 0.$$

They therefore cannot be independent without conditioning. Whether the system is working affects both alarms: learning that one monitor gave an alarm increases the probability that

the system is not working, which also increases the probability that the other monitor gave an alarm. Thus the alarms can be independent conditional on the system's status but dependent overall.

Marking guide: 2 marks for the joint and marginal expectations; 2 for the covariance identity; 1 for the independence conclusion; 1 for the explanation in words.

- (e) Let B^c denote the event that the second alarm does not sound. Derive $P(W = 0 \mid A \cap B^c)$ and the necessary and sufficient condition under which this probability is greater than its prior value π , and prove that it is less than $P(W = 0 \mid A)$. Explain the two comparisons in words.

[5 marks]

Solution : Conditional independence and Bayes' rule give

$$P(W = 0 \mid A \cap B^c) = \frac{\pi a_0(1 - b_0)}{\pi a_0(1 - b_0) + (1 - \pi)a_1(1 - b_1)}.$$

Since all probabilities are strictly between zero and one, direct cross-multiplication gives

$$P(W = 0 \mid A \cap B^c) > \pi \iff a_0(1 - b_0) > a_1(1 - b_1).$$

Also, comparing the displayed probability with the expression from part (a) shows that

$$P(W = 0 \mid A \cap B^c) < P(W = 0 \mid A) \iff 1 - b_0 < 1 - b_1,$$

which holds because $b_0 > b_1$. Alarm A is evidence that the system is not working, whereas not observing alarm B is evidence in the opposite direction. The first signal can still be stronger than the second.

Marking guide: 2 marks for the conditional probability; 1 for each comparison; 1 for interpreting the two signals.

Question 2

A continuous random variable X has density

$$f_X(x) = 3x^2, \quad 0 < x < 1,$$

and density zero otherwise. Define the odds transformation

$$Y = \frac{X}{1 - X}.$$

[25 marks]

- (a) Verify that f_X is a density and derive the cumulative distribution function (CDF) of X .

[4 marks]

Solution: The density is non-negative and $\int_0^1 3x^2 dx = 1$. Its CDF is

$$F_X(x) = \begin{cases} 0, & x \leq 0, \\ x^3, & 0 < x < 1, \\ 1, & x \geq 1. \end{cases}$$

Marking guide: 1 mark for non-negativity and unit integral; 2 for the interior CDF; 1 for the correct tail pieces and boundary behaviour.

- (b) State the possible values of Y (its support). Show that the probability density of Y is given by

$$f_Y(y) = \frac{3y^2}{(1+y)^4}.$$

[6 marks]

Solution: The odds transformation is increasing and maps $(0, 1)$ onto $(0, \infty)$. Its inverse and derivative are

$$x = \frac{y}{1+y}, \quad \frac{dx}{dy} = \frac{1}{(1+y)^2} > 0,$$

so the odds function is increasing. Hence, for $y > 0$,

$$F_Y(y) = P\left(X \leq \frac{y}{1+y}\right) = \left(\frac{y}{1+y}\right)^3$$

and

$$f_Y(y) = 3 \left(\frac{y}{1+y}\right)^2 \frac{1}{(1+y)^2} = \frac{3y^2}{(1+y)^4}.$$

The CDF is zero and the density is zero for $y \leq 0$.

Marking guide: 1 mark for the support and inverse; 1 for the derivative/Jacobian; 1 for the transformed CDF; 2 for the density; 1 for the values outside the support.

- (c) For $q \in (0, 1)$, derive the q th quantile of Y . State the median as a special case. **[4 marks]**

Solution: If y_q is the q th quantile, then

$$\left(\frac{y_q}{1 + y_q} \right)^3 = q.$$

Thus $y_q/(1 + y_q) = q^{1/3}$ and

$$y_q = \frac{q^{1/3}}{1 - q^{1/3}}.$$

In particular, the median is $y_{1/2} = 1/(2^{1/3} - 1)$.

Marking guide: 1 mark for the quantile equation; 2 for the general quantile; 1 for the median.

- (d) Use the CDF of Y to calculate $P(Y > 1)$ and $P(1 < Y < 3)$. **[5 marks]**

Solution: From part (b), for $y > 0$,

$$F_Y(y) = \left(\frac{y}{1 + y} \right)^3.$$

Therefore

$$P(Y > 1) = 1 - F_Y(1) = 1 - \frac{1}{8} = \frac{7}{8},$$

and

$$P(1 < Y < 3) = F_Y(3) - F_Y(1) = \frac{27}{64} - \frac{1}{8} = \frac{19}{64}.$$

Marking guide: 2 marks for $P(Y > 1)$; 3 for $P(1 < Y < 3)$, including the correct use of the CDF at both endpoints.

- (e) Calculate the values of $r > 0$ for which $\mathbb{E}[Y^r]$ is finite.

Hint: First write the expectation as an integral from 0 to 1, plus an integral from 1 to ∞ . For $y \geq 1$, use $y \leq 1 + y \leq 2y$ to compare the integrand with y^{r-2} . You may use the fact that $\int_1^\infty y^p dy$ is finite if and only if $p < -1$. **[6 marks]**

Solution: For $0 < y \leq 1$,

$$0 \leq \frac{3y^{r+2}}{(1+y)^4} \leq 3y^{r+2}.$$

Because $r > 0$, the integral of $3y^{r+2}$ from 0 to 1 is finite. Thus the part of the moment near zero is finite for every $r > 0$.

For $y \geq 1$, we have $y \leq 1 + y \leq 2y$, so $\frac{3}{16}y^{r-2} \leq 3y^{r+2}/(1+y)^4 \leq 3y^{r-2}$. Thus the upper part of the moment is finite exactly when $\int_1^\infty y^{r-2} dy$ is finite. The fact supplied in the question shows that this requires $r - 2 < -1$, or $r < 1$. Therefore

$$\mathbb{E}[Y^r] < \infty \iff 0 < r < 1.$$

Marking guide: 2 marks for showing that the integral from zero to one is finite; 2 for using or explaining the bounds for $y \geq 1$; 2 for applying the supplied power-integral rule and obtaining $0 < r < 1$.

Question 3

Let (X, Y) be distributed with joint density equal to

$$f_{X,Y}(x, y) = e^{-y}, \quad 0 < x < y < \infty,$$

and equal to zero otherwise.

[25 marks]

- (a) Show that the density is non-negative and that its integral over the support is equal to one. What are the marginal densities $f_X(x)$ and $f_Y(y)$?

[6 marks]

Solution: The function is non-negative and

$$\int_0^\infty \int_0^y e^{-y} dx dy = \int_0^\infty ye^{-y} dy = 1.$$

For $x > 0$,

$$f_X(x) = \int_x^\infty e^{-y} dy = e^{-x}.$$

For $y > 0$,

$$f_Y(y) = \int_0^y e^{-y} dx = ye^{-y}.$$

Both marginal densities are zero outside their stated positive supports.

Marking guide: 1 mark for verifying non-negativity and unit mass; 2 for f_X ; 2 for f_Y ; 1 for stating supports.

- (b) Derive $f_{X|Y}(x | y)$. What standard distribution does $X | Y = y$ have? Calculate $\mathbb{E}[X | Y = y]$ and $\text{Var}(X | Y = y)$.

[5 marks]

Solution: For $y > 0$,

$$f_{X|Y}(x | y) = \frac{e^{-y}}{ye^{-y}} = \frac{1}{y}, \quad 0 < x < y.$$

Thus $X | Y = y \sim U(0, y)$, so

$$\mathbb{E}[X | Y = y] = \frac{y}{2}, \quad \text{Var}(X | Y = y) = \frac{y^2}{12}.$$

Marking guide: 2 marks for the conditional density and support; 1 for identifying the uniform distribution; 2 for its conditional mean and variance.

- (c) From part (a), $Y \sim \text{Gamma}(2, 1)$, with $\mathbb{E}[Y] = 2$ and $\text{Var}(Y) = 2$. Use the law of iterated expectations and the law of total variance, together with the conditional moments in part (b), to calculate $\mathbb{E}[X]$ and $\text{Var}(X)$. **[4 marks]**

Solution: First,

$$\mathbb{E}[Y^2] = \text{Var}(Y) + \mathbb{E}[Y]^2 = 2 + 2^2 = 6.$$

The law of iterated expectations gives

$$\mathbb{E}[X] = \mathbb{E}[\mathbb{E}[X | Y]] = \frac{1}{2}\mathbb{E}[Y] = 1.$$

The law of total variance gives

$$\begin{aligned}\text{Var}(X) &= \mathbb{E}[\text{Var}(X | Y)] + \text{Var}(\mathbb{E}[X | Y]) \\ &= \frac{1}{12}\mathbb{E}[Y^2] + \frac{1}{4}\text{Var}(Y) \\ &= \frac{1}{2} + \frac{1}{2} = 1.\end{aligned}$$

Marking guide: 1 mark for obtaining $\mathbb{E}[Y^2] = 6$; 1 for applying iterated expectations and obtaining $\mathbb{E}[X] = 1$; 2 for correctly applying total variance and obtaining $\text{Var}(X) = 1$.

- (d) Calculate $\text{Cov}(X, Y)$. Are X and Y independent? Justify your answer. **[4 marks]**

Solution: By conditioning,

$$\mathbb{E}[XY] = \mathbb{E}[Y\mathbb{E}[X | Y]] = \frac{1}{2}\mathbb{E}[Y^2] = 3.$$

Since $\mathbb{E}[X] = 1$ and $\mathbb{E}[Y] = 2$,

$$\text{Cov}(X, Y) = 3 - (1)(2) = 1.$$

They are not independent, as the covariance is not 0. Alternatively, the support restriction $x < y$ and the conditional distribution $U(0, y)$ both show directly that the distribution of X changes with Y .

Marking guide: 1 mark for $\mathbb{E}[XY]$; 1 for the covariance; 2 for any valid non-independence argument.

- (e) Derive $f_{Y|X}(y | x)$. Define the gap $G = Y - X$ and use this conditional density to find the distribution of $G | X = x$. Hence determine whether X and G are independent. **[6 marks]**

Solution: For $x > 0$ and $y > x$,

$$f_{Y|X}(y | x) = \frac{f_{X,Y}(x, y)}{f_X(x)} = \frac{e^{-y}}{e^{-x}} = e^{-(y-x)}.$$

Consequently, for every $x > 0$,

$$f_{G|X}(g | x) = e^{-g}, \quad g > 0.$$

The conditional density and its support do not depend on x , so G is independent of X . Equivalently, for $x, g > 0$,

$$f_{X,G}(x, g) = f_X(x)f_{G|X}(g | x) = e^{-x}e^{-g},$$

which factorizes into the two marginal densities. It is not needed for the marks, but note that $Y = X + G$ is the sum of two independent rate-one exponential variables, which explains the $\text{Gamma}(2, 1)$ distribution found for Y .

Marking guide: 1 mark for $f_{Y|X}$ and its support; 1 for translating to the gap G ; 1 for the conditional exponential density; 1 for its distribution; 2 for independence.

Question 4

Consider the linear regression model

$$y = X\beta + \varepsilon, \quad \mathbb{E}[\varepsilon | X] = 0, \quad \hat{\beta} = (X'X)^{-1}X'y,$$

where X has full column rank. The observations are independent conditional on X , but their error variances may differ:

$$\Omega = \text{Var}(\varepsilon | X) = \text{diag}(\sigma_1^2, \dots, \sigma_n^2).$$

Here $\text{diag}(\sigma_1^2, \dots, \sigma_n^2)$ denotes the $n \times n$ diagonal matrix with diagonal entries $\sigma_1^2, \dots, \sigma_n^2$. An economist wants to test a scalar restriction

$$H_0 : a'\beta = \theta_0 \quad \text{against} \quad H_1 : a'\beta \neq \theta_0, \quad \hat{\theta} = a'\hat{\beta},$$

where a is a fixed non-zero vector.

[25 marks]

(a) Show that $\mathbb{E}[\hat{\beta} | X] = \beta$. Make sure to point out where you are using any assumptions.

[6 marks]

Solution: Substituting the regression model into the OLS formula gives

$$\hat{\beta} - \beta = (X'X)^{-1}X'\varepsilon.$$

Conditional zero mean therefore implies

$$\mathbb{E}[\hat{\beta} - \beta | X] = (X'X)^{-1}X'\mathbb{E}[\varepsilon | X] = 0,$$

so $\mathbb{E}[\hat{\beta} | X] = \beta$. The result follows from the conditional zero-mean assumption, $\mathbb{E}[\varepsilon | X] = 0$.

Marking guide: 2 marks for writing $\hat{\beta} - \beta$ in terms of ε ; 3 for taking the conditional expectation; 1 for identifying the conditional zero-mean assumption.

(b) Derive

$$\text{Var}(\hat{\beta} | X) = (X'X)^{-1}X'\Omega X(X'X)^{-1}.$$

Show how this formula simplifies under homoscedasticity.

[6 marks]

Solution: From part (a),

$$\hat{\beta} - \beta = (X'X)^{-1}X'\varepsilon.$$

Taking the conditional variance gives

$$\text{Var}(\hat{\beta} | X) = (X'X)^{-1}X'\Omega X(X'X)^{-1}.$$

Under homoscedasticity, $\Omega = \sigma^2 I_n$, so

$$\text{Var}(\hat{\beta} | X) = \sigma^2 (X'X)^{-1}.$$

Marking guide: 1 mark for using the expression from part (a); 3 for the general conditional variance formula; 2 for the homoscedastic simplification.

- (c) Use part (b) to derive $\text{Var}(\hat{\theta} \mid X)$. Let $\widehat{\text{se}}_{\text{White}}(\hat{\theta})$ denote its estimated White standard error. Write down the corresponding test statistic for testing $H_0 : a'\beta = \theta_0$ and describe when we would reject the test. What pattern of heteroscedasticity do White standard errors allow for? Are they valid in the presence of autocorrelation? **[8 marks]**

Solution: Writing $V = \text{Var}(\hat{\beta} \mid X)$, the variance of the scalar contrast is

$$\text{Var}(\hat{\theta} \mid X) = \text{Var}(a'\hat{\beta} \mid X) = a'Va = a'(X'X)^{-1}X'\Omega X(X'X)^{-1}a.$$

White standard errors allow the conditional variance to differ across observations in any way. The ordinary White formula is not valid under autocorrelation, because it does not include off-diagonal error covariances.

The large-sample test statistic is

$$T = \frac{\hat{\theta} - \theta_0}{\widehat{\text{se}}_{\text{White}}(\hat{\theta})}.$$

Writing $z_{1-\alpha/2}$ for the relevant standard-Normal quantile, reject the null when $|T| > z_{1-\alpha/2}$. This rule is justified asymptotically rather than as an exact finite-sample result under the stated assumptions.

Marking guide: 2 marks for the scalar-contrast variance; 2 for the test statistic; 1 for the two-sided critical-value rule (or something alternative like a p value or critical value) and its asymptotic nature; 3 for the scope of White standard errors and the autocorrelation qualification.

- (d) Explain the standard trade-off between type I and type II errors. Why does lowering the significance level α make a type I error less likely but a type II error more likely? **[5 marks]**

Solution: For a level- α test, the probability of rejecting the null when it is true is at most α . Lowering α therefore reduces the permitted probability of a type I error. For the usual test with nested rejection regions, a lower α uses a more demanding critical value and hence a smaller rejection region. At a fixed alternative, rejection is therefore less likely: power falls and the probability of failing to reject a false null, the type II error probability, rises. This is the standard trade-off between the two error probabilities.

Marking guide: 2 marks for the effect on type I error; 2 for linking the smaller rejection region to lower power and higher type II error; 1 for a clear statement of the trade-off. Full marks awarded if the answer is essentially a restatement of the legal analogy from lecture.

Question 5

A university invites students to volunteer for a study of a new online revision programme, then selects 150 students at random from the population of volunteers. Within this group, 90 students are randomly assigned access to the programme ($D = 1$), while 60 are assigned to the control group ($D = 0$). Let Y be the end-of-term score. For the population of volunteers, consider the regression

$$Y = \alpha + \beta D + \varepsilon, \quad \mathbb{E}[\varepsilon \mid D] = 0.$$

The realized mean score is 78 in the programme group and 71 in the control group.

[25 marks]

- (a) Use $\mathbb{E}[\varepsilon \mid D] = 0$ to show that

$$\alpha = \mathbb{E}[Y \mid D = 0] \quad \text{and} \quad \beta = \mathbb{E}[Y \mid D = 1] - \mathbb{E}[Y \mid D = 0].$$

[5 marks]

Solution: When $D = 0$, taking conditional expectations gives

$$\mathbb{E}[Y \mid D = 0] = \alpha + \mathbb{E}[\varepsilon \mid D = 0] = \alpha.$$

When $D = 1$,

$$\mathbb{E}[Y \mid D = 1] = \alpha + \beta + \mathbb{E}[\varepsilon \mid D = 1] = \alpha + \beta.$$

Subtracting the first equality from the second gives the stated expression for β .

Marking guide: 2 marks for conditioning at $D = 0$; 2 for conditioning at $D = 1$; 1 for subtracting to obtain the expression for β .

- (b) In the regression above, calculate $\hat{\alpha}$ and $\hat{\beta}$. Briefly explain what $\hat{\beta}$ compares. [4 marks]

Solution: With an intercept and a binary regressor, OLS fits the two group means. Hence

$$\hat{\alpha} = \bar{Y}_0 = 71, \quad \hat{\beta} = \bar{Y}_1 - \bar{Y}_0 = 78 - 71 = 7.$$

Thus $\hat{\beta}$ compares the mean score in the programme group with the mean score in the control group. The realized programme-group mean is 7 points higher.

Marking guide: 1 mark for $\hat{\alpha}$; 2 for $\hat{\beta}$; 1 for the comparison in words.

- (c) For a student i , let $Y_i(1)$ denote what the score would have been if the student had received access to the programme, and let $Y_i(0)$ be the score if they had not received it.
- Write the observed outcome Y_i in terms of D_i , $Y_i(1)$ and $Y_i(0)$.
 - State the random-assignment condition using D_i , $Y_i(1)$ and $Y_i(0)$.
 - Use random sampling and random assignment to show that

$$\beta = \mathbb{E}[Y(1) - Y(0)],$$

where the expectation is for the population of volunteers.

[7 marks]

Solution: The observed outcome is

$$Y_i = D_i Y_i(1) + (1 - D_i) Y_i(0).$$

Within the randomly selected study sample, random assignment means that

$$D_i \perp\!\!\!\perp (Y_i(0), Y_i(1)).$$

Using the observed-outcome equation and random assignment, the treatment-group means identify the corresponding potential-outcome means for the study participants. Because the participants were randomly sampled from the population of volunteers, these potential-outcome means in turn identify the corresponding means for that population. Thus

$$\mathbb{E}[Y \mid D = 1] = \mathbb{E}[Y(1)] \quad \text{and} \quad \mathbb{E}[Y \mid D = 0] = \mathbb{E}[Y(0)].$$

Therefore

$$\beta = \mathbb{E}[Y(1)] - \mathbb{E}[Y(0)] = \mathbb{E}[Y(1) - Y(0)].$$

Marking guide: 2 marks for the observed-outcome equation; 2 for the independence condition; 3 for showing that β equals the population ATE.

- (d) Does random assignment make the 150 student volunteers representative of all students at the university? Explain the different roles of random assignment and random sampling.

[5 marks]

Solution : No. Random assignment determines which study participants receive access to the programme. It makes the two assigned groups comparable, supporting a causal interpretation for the study participants. Random sampling determines which volunteers enter the study and supports generalization from the study participants to the population of volunteers. Random selection from the population of volunteers does not make that population representative of all students at the university.

Marking guide: 1 mark for answering no; 2 for the role of random assignment; 2 for the role of random sampling.

- (e) Now suppose instead that the university takes a random sample of 150 students, but each student chooses whether to use the online revision programme. What would random sampling achieve? Why would the difference in mean scores not automatically have a causal interpretation?

[4 marks]

Solution : Random sampling would make the study students representative of the university's students under the sampling design. It would therefore support generalization of descriptive quantities to that population.

However, students choosing the programme may differ from those who do not, for example in prior attainment or motivation. Thus D may be related to $(Y(0), Y(1))$. The difference in mean scores could then reflect selection as well as the effect of the programme, so it would not automatically identify an average treatment effect.

Marking guide: 2 marks for what random sampling achieves; 1 for explaining self-selection; 1 for explaining why this prevents an automatic causal interpretation.

Question 6

Let n be even. The independent observations $X_1, \dots, X_n$ all have mean μ and are Normal, but their variances alternate between two positive constants:

$$\text{Var}(X_i) = \begin{cases} v_L, & i \text{ odd}, \\ v_H, & i \text{ even}, \end{cases} \quad 0 < v_L < v_H.$$

Assume that v_L and v_H are known. For $w > 0$, consider the weighted mean

$$T_n(w) = \frac{\sum_{i \text{ odd}} X_i + w \sum_{i \text{ even}} X_i}{(n/2)(1+w)}.$$

Thus each group contains $n/2$ observations. The odd-indexed observations form the lower-variance group, while w is the weight on each observation in the higher-variance group relative to an odd-indexed observation. The denominator normalizes the weights so that they sum to one.

[25 marks]

- (a) For arbitrary $w > 0$, calculate $\mathbb{E}[T_n(w)]$ and show that $T_n(w)$ is an unbiased estimator of μ .

[4 marks]

Solution: The expected numerator is $(n/2)(1+w)\mu$, so

$$\mathbb{E}[T_n(w)] = \frac{(n/2)(1+w)\mu}{(n/2)(1+w)} = \mu.$$

Therefore $T_n(w)$ is unbiased for every $w > 0$.

Marking guide: 2 marks for the expected numerator; 1 for dividing by the denominator; 1 for concluding that the estimator is unbiased.

- (b) For arbitrary $w > 0$, let

$$S_n(w) = \sum_{i \text{ odd}} X_i + w \sum_{i \text{ even}} X_i$$

denote the numerator of $T_n(w)$. Show that $\text{Var}[S_n(w)] = (n/2)(v_L + w^2 v_H)$. Hence derive the variance of $T_n(w)$. Explain why $T_n(w)$ is Normally distributed and state its exact distribution.

[6 marks]

Solution: Independence gives the variance of the numerator:

$$\frac{n}{2}(v_L + w^2 v_H).$$

After dividing by the square of the denominator,

$$\text{Var}[T_n(w)] = \frac{\frac{n}{2}(v_L + w^2 v_H)}{[(n/2)(1+w)]^2} = \frac{2(v_L + w^2 v_H)}{n(1+w)^2}.$$

Because $T_n(w)$ is a linear combination of independent Normal random variables, it is also Normal. Using its mean from part (a),

$$T_n(w) \sim N\left(\mu, \frac{2(v_L + w^2 v_H)}{n(1+w)^2}\right).$$

Marking guide: 2 marks for the numerator variance; 2 for scaling and simplifying the variance; 1 for explaining Normality; 1 for the exact distribution.

- (c) Show that the value of w that minimizes the variance of $T_n(w)$ is $w^* = v_L/v_H$. You may use $\frac{d}{dw} \{(v_L + w^2 v_H)/(1 + w)^2\} = 2(w v_H - v_L)/(1 + w)^3$. Verify that this is a minimum, and give some intuition for why the higher-variance observations receive less weight. **[6 marks]**

Solution: Apart from the positive factor $2/n$, the derivative of the variance factor is

$$\frac{d}{dw} \frac{v_L + w^2 v_H}{(1 + w)^2} = \frac{2(w v_H - v_L)}{(1 + w)^3}.$$

The derivative is negative for $w < v_L/v_H$ and positive for $w > v_L/v_H$. Hence the unique minimizer is

$$w^* = \frac{v_L}{v_H}.$$

Since $v_H > v_L$, the even observations are noisier and $w^* = v_L/v_H < 1$ gives them less weight. This reduces their contribution to the estimator's variance. At the optimum, their relative weight is

$$\frac{v_L}{v_H} = \frac{1/v_H}{1/v_L},$$

so observations receive weights proportional to their precision, the inverse of their variance.

Marking guide: 2 marks for using the derivative to locate $w^* = v_L/v_H$; 2 for verifying that it is the minimum; 2 for the intuition about giving less weight to noisier observations.

- (d) Let $\bar{X}_n = n^{-1} \sum_{i=1}^n X_i$. Notice that $\bar{X}_n = T_n(1)$. You may use $\text{Var}(\bar{X}_n) = (v_L + v_H)/(2n)$. Use parts (b) and (c) to show that $\text{Var}(\bar{X}_n) - \text{Var}[T_n(w^*)] = (v_H - v_L)^2/[2n(v_L + v_H)]$. Hence show that the optimally weighted estimator has strictly smaller variance under the maintained assumption $v_L < v_H$. If equal variances were allowed, when would the two estimators have the same variance? Give an intuitive explanation. **[5 marks]**

Solution: There are $n/2$ observations of each variance, so

$$\text{Var}(\bar{X}_n) = \frac{v_L + v_H}{2n}.$$

Substitution of $w^* = v_L/v_H$ gives

$$\text{Var}[T_n(w^*)] = \frac{2v_L v_H}{n(v_L + v_H)}.$$

The reduction in variance is

$$\text{Var}(\bar{X}_n) - \text{Var}[T_n(w^*)] = \frac{(v_H - v_L)^2}{2n(v_L + v_H)} \geq 0.$$

Under the maintained assumption $v_L < v_H$, the difference is strictly positive. If equal variances were allowed, equality would hold exactly when $v_L = v_H$. If all observations are equally informative, equal weights are efficient within this class; when their noise levels differ, down-weighting noisier observations strictly reduces variance without introducing bias.

Marking guide: 2 marks for simplifying the difference; 1 for showing it is non-negative and giving the equality condition; 2 for the interpretation.

- (e) For a fixed $w > 0$, use the mean and variance from parts (a) and (b) to show that $T_n(w) \xrightarrow{p} \mu$. Then state the exact distribution of $\sqrt{n}[T_n(w) - \mu]$, and hence its asymptotic variance. What is this asymptotic variance at $w = w^*$? **[4 marks]**

Solution : For every $\epsilon > 0$, Chebyshev's inequality and parts (a) and (b) give

$$\Pr(|T_n(w) - \mu| > \epsilon) \leq \frac{2(v_L + w^2 v_H)}{n(1 + w)^2 \epsilon^2} \rightarrow 0.$$

Thus $T_n(w) \xrightarrow{p} \mu$. Its exact Normal distribution from part (b) implies

$$\sqrt{n}[T_n(w) - \mu] \sim N\left(0, \frac{2(v_L + w^2 v_H)}{(1 + w)^2}\right).$$

The expression on the right is both its exact variance for every even n and its asymptotic variance. At $w^* = v_L/v_H$, it becomes

$$\frac{2v_L v_H}{v_L + v_H}.$$

Marking guide: 2 marks for the consistency argument; 1 for the root- n distribution and general asymptotic variance; 1 for the asymptotic variance at w^* .